

Student's Name: _____ Teacher's Code : _____



Saint Ignatius' College, Riverview

Mathematics Assessment Task

2022

Year 12
Mathematics Extension 1
Task 1
In-Class Assessment
Date: 30 November 2021

General Instructions:	Topics Examined:																
• Time Allowed: 45 minutes • Write using blue or black pen only • Board approved calculators may be used • Attempt all questions in the space provided on the paper • Write your name and your teacher's code in the positions indicated • Marks may be deducted for missing or carelessly arranged working	<table><tr><td>Question 1</td><td>10 marks</td></tr><tr><td>Polynomials</td><td></td></tr><tr><td>Question 2</td><td>10 marks</td></tr><tr><td>Polynomials</td><td></td></tr><tr><td>Question 3</td><td>10 marks</td></tr><tr><td>Mathematical Induction</td><td></td></tr><tr><td colspan="2"> </td></tr><tr><td>Total</td><td>30 Marks</td></tr></table>	Question 1	10 marks	Polynomials		Question 2	10 marks	Polynomials		Question 3	10 marks	Mathematical Induction				Total	30 Marks
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Teachers : • Mr R Maxwell REM • Mr D Reidy DPR • Mr N Mushan NHM • Mr J Newey JPN																	

Question 1 (10 marks). Use a SEPARATE writing booklet.

Marks

- (a) The quadratic equation $4x^2 - px - 5 = 0$ has two roots opposite in sign but equal in magnitude. Find the value of p .

2

- (b) Find the values of m and n if $(x-1)$ and $(x+2)$ are factors of

$$P(x) = x^4 + 4x^3 + mx^2 - n$$

2

- (c) i) Show that $P(x) = 2x^3 + x^2 - 13x + 6$ is divisible by $(2x-1)$ and $(x+3)$

2

- ii) Hence, find all the **zeros** of $P(x)$

1

- (d) Given that the polynomial $x^2 + x + 2$ is a factor of $x^4 + 4x^2 + x + a$.

- i) Using the process of Long Division of Polynomials, or otherwise, find the other factor.

2

- ii) Hence, find the value of a .

1

Question 2 (10 marks). Use a SEPARATE writing booklet.

Marks

- (a) Solve the equation $4x^3 + 32x^2 + 79x + 60 = 0$ given that one root is the sum of the other two roots.

3

2

- (b) Consider the function $f(x) = x^3 - x^2 - 8x + 12$.

- i) Show that the graph of $y = f(x)$ cuts the x -axis at one point and touches it at another.

2

- ii) Sketch the graph of $y = f(x)$ clearly identifying all intercepts.

2

- (c) The roots of $x^3 - x^2 - 5x + 2 = 0$ are α , β and γ

- i) Show that $\beta + \gamma = 1 - \alpha$

1

- ii) Using part (i), and similar results, evaluate $\frac{\beta + \gamma}{\alpha} + \frac{\gamma + \alpha}{\beta} + \frac{\alpha + \beta}{\gamma}$

2

Question 3 (10 marks) Use a SEPARATE writing booklet

Marks

- (a) Prove by Mathematical Induction that for all positive integers $n \geq 1$

3

$$1 \cdot 2 \cdot 3 + 2 \cdot 3 \cdot 4 + 3 \cdot 4 \cdot 5 + \dots + n(n+1)(n+2) = \frac{1}{4}n(n+1)(n+2)(n+3)$$

- (b) Prove by Mathematical Induction that for all positive integers $n \geq 1$

$$7^{2n} + 7^n + 4 \text{ is divisible by } 6.$$

3

- (c) (i) Use Mathematical Induction to prove for all integers $n \geq 1$, that

$$\left(\frac{1 \times 5}{3^2}\right)\left(\frac{3 \times 7}{5^2}\right)\left(\frac{5 \times 9}{7^2}\right) \cdots \left(\frac{(2n+1)(2n+3)}{(2n+1)^2}\right) = \frac{2n+3}{6n+3}$$

3

- (ii) As n approaches infinity (becomes very large), what does the value of the following expression approach

$$\left(\frac{1 \times 5}{3^2}\right)\left(\frac{3 \times 7}{5^2}\right)\left(\frac{5 \times 9}{7^2}\right) \cdots$$

1

END OF ASSESSMENT TASK

Student's Name: _____ Teacher's Code : _____



SUGGESTED SOLUTIONS

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$$(a) \quad 4x^2 - px - 5 = 0$$

$$(2x - \sqrt{5})(2x + \sqrt{5}) = 0$$

$$\therefore p = 0.$$

$$(b) \quad P(x) = x^4 + 4x^3 + mx^2 - n$$

$$P(1) = 1 + 4 + m - n = 0$$

$$5 + m - n = 0 \quad \textcircled{1}$$

$$P(-2) = 16 - 32 + 4m - n = 0$$

$$-16 + 4m - n = 0 \quad \textcircled{2}$$

$$\textcircled{2} - \textcircled{1}$$

$$-21 + 3m = 0$$

$$m = 7$$

$$\text{Sub into } \textcircled{1} \quad 5 + 7 - n = 0$$

$$n = 12$$

$$(c) \quad P(x) = 2x^3 + x^2 - 13x + 6$$

$$2x - 1 = 0$$

$$x = \frac{1}{2}$$

$$P\left(\frac{1}{2}\right) = 2\left(\frac{1}{2}\right)^3 + \left(\frac{1}{2}\right)^2 - 13\left(\frac{1}{2}\right) + 6$$

$$= \frac{1}{4} + \frac{1}{4} - \frac{13}{2} + 6$$

$$= 0 \quad \therefore (2x - 1) \text{ is a factor}$$

$$P(-3) = 2(-3)^3 + (-3)^2 - 13(-3) + 6$$

$$= -54 + 9 + 39 + 6$$

$$= 0 \quad \therefore (x + 3) \text{ is a factor}$$

Let $(x + a)$ be the third factor

$$\therefore -1 \times 3 \times a = 6$$

$$a = -2, \text{ zeros: } \frac{1}{2}, -3, 2.$$

$$\begin{array}{r}
 (d)(i) \quad x^2 - x + 3 \\
 x^2 + x + 2 \overline{) x^4 + 4x^2 + x + a} \\
 \underline{x^4 + x^3 + 2x^2 -} \\
 -x^3 + 2x^2 + x \\
 \underline{-x^3 - x^2 - 2x} \\
 3x^2 + 3x + a - \\
 \underline{3x^2 + 3x + 6}
 \end{array}$$

$$\begin{aligned}
 (ii) \quad r &= a - b = 0 \\
 \therefore \underline{a} &= \underline{b}
 \end{aligned}$$

$$Q2. (a) 4x^3 + 32x^2 + 79x + 60 = 0$$

Let the roots be α , β and $(\alpha + \beta)$

$$\alpha + \beta + (\alpha + \beta) = -\frac{b}{a}$$

$$2\alpha + 2\beta = -\frac{32}{4}$$

$$2(\alpha + \beta) = -8$$

$$\alpha + \beta = -4$$

$$\begin{aligned}
 \alpha \cdot \beta \cdot (\alpha + \beta) &= \frac{-d}{a} \\
 &= \frac{-60}{4} \\
 &= -15
 \end{aligned}$$

$$\therefore \alpha \beta (-4) = -15$$

$$\alpha \beta = \frac{15}{4}$$

$$\alpha + \beta = -4$$

$$\therefore \beta = -4 - \alpha$$

(a) CONTINUED

$$\alpha + \beta = \frac{15}{4}$$

$$x(-4-x) = \frac{15}{4}$$

$$-4x - x^2 = \frac{15}{4}$$

$$\therefore x^2 + 4x + \frac{15}{4} = 0$$

$$\therefore x = \frac{-4 \pm \sqrt{16 - 4 \cdot \frac{15}{4}}}{2}$$

$$= \frac{-4 \pm 1}{2}$$

$$= -\frac{3}{2}, -\frac{5}{2}$$

$$\therefore \alpha + \beta = -4$$

$$\text{if } \alpha = -\frac{3}{2},$$

$$\beta = -4 + \frac{3}{2}$$

$$= -\frac{5}{2}$$

roots are: $-4, -\frac{5}{2}, -\frac{3}{2}$

$$(b) f(x) = x^3 - x^2 - 8x + 12$$

By inspection $f(x) = (x+3)(x-2)(x-2)$
check:

$$f(2) = 2^3 - 2^2 - 8(2) + 12$$

$$= 8 - 4 - 16 + 12$$

$$= 0$$

$$f(-3) = (-3)^3 - (-3)^2 - 8(-3) + 12$$

$$= -27 - 9 + 24 + 12$$

$$= 0$$

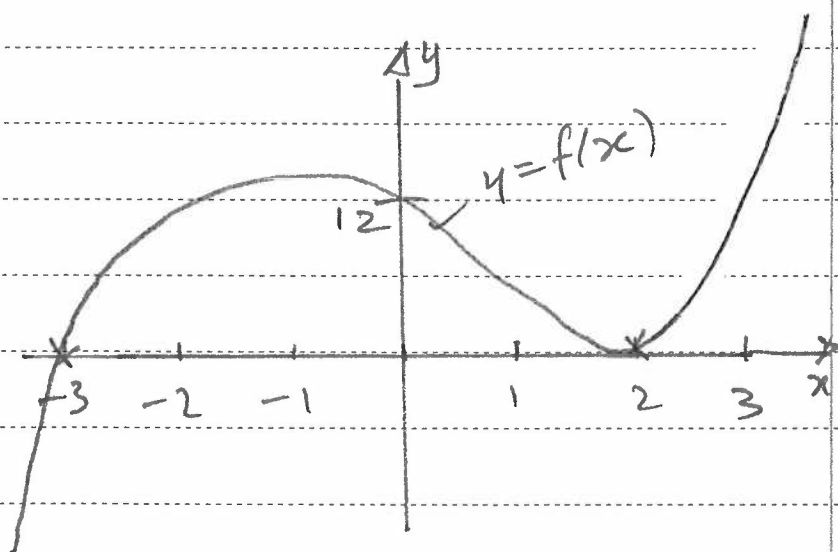
$$f(x) = (x-2)(x+3)(x+a)$$

$$-2 \times 3 \times a = 12$$

$$a = -2$$

$\therefore (x-2)$ is a repeated factor.

(b) continued



Mark 1: Correct shape and position
Mark 2: Three intercepts shown

(c) $x^3 - x^2 - 5x + 2 = 0$

(i) $\alpha + \beta + \gamma = \frac{-(-1)}{1}$
 $= 1$

$\therefore \alpha + \beta = 1 - \gamma$

(ii) Similarly $\alpha + \gamma = 1 - \beta$, $\beta + \gamma = 1 - \alpha$

$\therefore \frac{\beta + \gamma}{\alpha} + \frac{\alpha + \gamma}{\beta} + \frac{\alpha + \beta}{\gamma} = \frac{1 - \alpha}{\alpha} + \frac{1 - \beta}{\beta} + \frac{1 - \gamma}{\gamma}$

$= \frac{\beta\gamma(1 - \alpha) + \alpha\gamma(1 - \beta) + \alpha\beta(1 - \gamma)}{\alpha\beta\gamma}$

$= \frac{\beta\gamma - \alpha\beta\gamma + \alpha\gamma - \alpha\beta\gamma + \alpha\beta - \alpha\beta\gamma}{\alpha\beta\gamma}$

$= \frac{\beta\gamma + \alpha\gamma + \alpha\beta - 3\alpha\beta\gamma}{\alpha\beta\gamma}$

$= \frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} - 3$

$= \frac{-5}{-2} - 3$

$= -\frac{1}{2}$

Q 3

(a) Prove $1 \cdot 2 \cdot 3 + 2 \cdot 3 \cdot 4 + 3 \cdot 4 \cdot 5 + \dots + n(n+1)(n+2) = \frac{1}{4} n(n+1)(n+2)(n+3)$

1. Prove true for $n=1$

$$\text{i.e. } n(n+1)(n+2) = \frac{1}{4} n(n+1)(n+2)(n+3)$$

$$\text{LHS} = 1 \times 2 \times 3$$

$$= 6$$

$$\text{RHS} = \frac{1}{4} \cdot 1(2)(3)(4)$$

$$= 6$$

$\therefore$ true for $n=1$.

2. Assume true for $n=k$

$$\text{i.e. } 1 \cdot 2 \cdot 3 + 2 \cdot 3 \cdot 4 + 3 \cdot 4 \cdot 5 + \dots + k(k+1)(k+2) = \frac{1}{4} k(k+1)(k+2)(k+3)$$

3. Prove true for $n=k+1$

$$\begin{aligned} \text{i.e. } 1 \cdot 2 \cdot 3 + 2 \cdot 3 \cdot 4 + 3 \cdot 4 \cdot 5 + \dots + k(k+1)(k+2) + (k+1)(k+2)(k+3) \\ = \frac{1}{4} (k+1)(k+2)(k+3)(k+4) \end{aligned}$$

$$\text{LHS} = 1 \cdot 2 \cdot 3 + 2 \cdot 3 \cdot 4 + 3 \cdot 4 \cdot 5 + \dots + k(k+1)(k+2) + (k+1)(k+2)(k+3)$$

$$= \frac{1}{4} k(k+1)(k+2)(k+3) + (k+1)(k+2)(k+3)$$

$$= \left(\frac{1}{4} k + 1 \right) (k+1)(k+2)(k+3)$$

$$= \frac{1}{4} (k+4)(k+1)(k+2)(k+3)$$

$$= \text{RHS}$$

$\therefore$ true for $k+1$ if true for k . As true for $n=1$, true for $n=2$ and so on. True for all $n \geq 1$.

(b) Prove $7^{2n} + 7^n + 4$ is divisible by 6, $n \geq 1$

1. Prove true for $n=1$

$$\begin{aligned} 7^2 + 7 + 4 &= 60 \\ &= 6 \times 10 \therefore \text{true for } n=1 \end{aligned}$$

2. Assume true for $n=k$

i.e. $7^{2k} + 7^k + 4 = 6M$, M is an integer

3. Prove true for $n=k+1$

i.e. $7^{2(k+1)} + 7^{k+1} + 4 = 6N$, N is an integer.

$$\begin{aligned} \text{LHS} &= 7^2 \cdot 7^{2k} + 7 \cdot 7^k + 4 \\ &= 7^2 [6M - 7^k - 4] + 7 \cdot 7^k + 4 \\ &= 6 \times 49M - 42 \cdot 7^k - 48 + 4 \\ &= 6 [49M - 7 \cdot 7^k - 32] \\ &= 6N \quad \left[\text{as } 49M, 7 \cdot 7^k \text{ and } 32 \right. \\ &\quad \left. \text{are all integers} \right] \end{aligned}$$

$\therefore$ True for $n=k+1$ if true for $n=k$
as true for $n=1$, true for $n=2$,
and so on... true for all $n \geq 1$.

$$(k) \text{ RTP } \left(\frac{1 \times 5}{3^2} \right) \left(\frac{3 \times 7}{5^2} \right) \left(\frac{5 \times 9}{7^2} \right) \dots \left[\frac{(2n-1)(2n+3)}{(2n+1)^2} \right] = \frac{2n+3}{6n+3}$$

1. Prove true for $n=1$.

$$\text{LHS} = \frac{(2-1)(2+3)}{(2+1)^2} \quad \text{RHS} = \frac{2+3}{6+3}$$

$$= \frac{5}{9} \quad = \frac{5}{9}$$

$\therefore$ true for $n=1$.

2. Assume true for $n=k$.

$$1 \times \left(\frac{1 \times 5}{3^2} \right) \left(\frac{3 \times 7}{5^2} \right) \left(\frac{5 \times 7}{7^2} \right) \dots \left[\frac{(2k-1)(2k+3)}{(2k+1)^2} \right] = \frac{2k+3}{6k+3}$$

3. Prove true for $n=k+1$

$$1 \times \left(\frac{1 \times 5}{3^2} \right) \left(\frac{3 \times 7}{5^2} \right) \left(\frac{5 \times 7}{7^2} \right) \dots \left[\frac{(2k-1)(2k+3)}{(2k+1)^2} \right] \left[\frac{(2k+1)(2k+5)}{(2k+3)^2} \right] = \frac{2k+5}{6k+9}$$

$$\text{LHS} = \left(\frac{1 \times 5}{3^2} \right) \left(\frac{3 \times 7}{5^2} \right) \dots \left[\frac{(2k-1)(2k+3)}{(2k+1)^2} \right] \left[\frac{(2k+1)(2k+5)}{(2k+3)^2} \right]$$

$$= \left[\frac{2k+3}{6k+3} \right] \left[\frac{(2k+1)(2k+5)}{(2k+3)^2} \right]$$

$$= \frac{(2k+1)(2k+5)}{3(2k+1)(2k+3)}$$

$$= \frac{2k+5}{6k+9}$$

$$= \frac{2k+5}{6k+9}$$

$$= \text{RHS}$$

$$= \text{RHS}$$

$\therefore$ true for $n=k+1$ if true for $n=k$.

as true for $n=1$, true for $n=2$

and so on $\dots \therefore$ true for all $n \geq 1$.

Q3 (c) (ii) [Formal limits not covered]

$$\left(\frac{1 \times 5}{3^2}\right) \left(\frac{3 \times 7}{5^2}\right) \left(\frac{5 \times 9}{7^2}\right) \dots = \frac{2n+3}{6n+3}$$

let $n = 10\,000$ say,

$$\therefore \frac{2(10\,000)+3}{6(10\,000)+3} = 0.33336.$$

$$\div \frac{1}{3}.$$

Seems to be heading to $\frac{1}{3}$.